Algorithm 10: Bounded Forward Half-Search

Input:

- V. Set of nodes
- E_s. Set of scheduled edges $(v, u, \text{dep}, \text{arr})$
- E_t. Set of transfer edges (v, w, Δ)
- $\text{country}(v)$. Country associated with node v
- $R(v, b)$. Precomputed reachable country sets (from Algorithm 3)
- $MTT(c)$. Precomputed minimum transit times (from Algorithm 5)
- $\text{bucket}(t)$. Time bucket mapping function
- MaxDuration. Maximum allowed elapsed time for the half-search (e.g., 840 mins / 14 hours)
- MinCountries. Minimum unique countries required to save the path (e.g., 6)
- GlobalMinTime. Minimum realistic time to visit a country (e.g., 25 mins)

Output:

- Labels_F. Map of nodes to a set of Pareto-optimal forward labels

Initialize $\text{Labels}_F(v) \leftarrow \emptyset$ for all $v \in V$
Initialize priority queue $Q \leftarrow \emptyset$ // ordered by decreasing score
// Initialize labels at departures
for all $v \in V$ **do**
  for all scheduled edges $(v, u, \text{dep}, \text{arr}) \in E_s$ **do**
    $L \leftarrow (\text{node} = u, t = \text{arr}, C = \{\text{country}(u)\}, \text{pred} = \text{null}, \text{start\_time} = \text{dep})$
    $\text{Labels}_F(u) \leftarrow \text{Labels}_F(u) \cup \{L\}$
    compute $\text{score}(L) \leftarrow (|L.C| \times 10000) - L.t$
    $Q \leftarrow Q \cup \{L\}$
// Main loop
while $Q \neq \emptyset$ **do**
  extract $L = (v, t, C, \text{pred}, \text{start\_time})$ with maximum score from Q
  $\text{elapsed} \leftarrow t - \text{start\_time}$
  $\text{remaining\_half\_time} \leftarrow \text{MaxDuration} - \text{elapsed}$
  // — 1. GLOBAL HARD FLOOR PRUNING (Half-Search Variant) —
  $\text{needed} \leftarrow \text{MinCountries} - |C|$
  // If the path physically cannot even reach the minimum required countries in the allowed 14 hours, kill it.
  if $\text{needed} > 0 \wedge \text{remaining\_half\_time} < (\text{needed} \times \text{GlobalMinTime})$ **then**
    continue
  // — 2. TIME-BUDGETED UB PRUNING (Half-Search Variant) —
  if $\text{needed} > 0$ **then**
    $b \leftarrow \text{bucket}(t)$
    $U \leftarrow R(v, b) \setminus C$
    Sort U as a sequence $(c_1, c_2, \ldots, c_k)$ such that $MTT(c_1) \leq MTT(c_2) \leq \ldots \leq MTT(c_k)$
    $\text{time\_spent} \leftarrow 0$
    $\text{UB} \leftarrow 0$
    for $i = 1$ to $|U|$ **do**
      $\text{time\_spent} \leftarrow \text{time\_spent} + MTT(c_i)$
      if $\text{time\_spent} \leq \text{remaining\_half\_time}$ **then**
        $\text{UB} \leftarrow \text{UB} + 1$
      else

break
// *If current countries + theoretically reachable countries ¡ minimum acceptable, kill it.*
if $|C| + \mathrm{UB} < \mathrm{MinCountries}$ **then**
  continue
// *— 3. EXTEND VIA SCHEDULED EDGES —*
for all $(v, u, \mathrm{dep}, \mathrm{arr}) \in E_s$ **do**
  if $\mathrm{dep} \geq t$ **then**
    $\mathrm{new\_elapsed} \leftarrow \mathrm{arr} - \mathrm{start\_time}$
    // *Enforce the 14-hour cut-off limit*
    if $\mathrm{new\_elapsed} \leq \mathrm{MaxDuration}$ **then**
      $C' \leftarrow C \cup \{\mathrm{country}(u)\}$
      $L' \leftarrow (u, \mathrm{arr}, C', \mathrm{pred} = L, \mathrm{start\_time} = \mathrm{start\_time})$

      // *Strict Pareto Dominance*
      if $\nexists L'' \in \mathrm{Labels}_F(u)$ such that
        $(L''.t \leq \mathrm{arr} \wedge L''.C \supseteq C' \wedge (L''.t < \mathrm{arr} \vee L''.C \neq C'))$ **then**
        remove all $L'' \in \mathrm{Labels}_F(u)$ such that
          $(\mathrm{arr} \leq L''.t \wedge C' \supseteq L''.C \wedge (\mathrm{arr} < L''.t \vee C' \neq L''.C))$
        $\mathrm{Labels}_F(u) \leftarrow \mathrm{Labels}_F(u) \cup \{L'\}$
        compute score(L')
        $Q \leftarrow Q \cup \{L'\}$
// *— 4. EXTEND VIA TRANSFERS —*
for all $(v, w, \Delta) \in E_t$ **do**
  $t_{\mathrm{new}} \leftarrow t + \Delta$
  $\mathrm{new\_elapsed} \leftarrow t_{\mathrm{new}} - \mathrm{start\_time}$
  if $\mathrm{new\_elapsed} \leq \mathrm{MaxDuration}$ **then**
    $L' \leftarrow (w, t_{\mathrm{new}}, C, \mathrm{pred} = L, \mathrm{start\_time} = \mathrm{start\_time})$

    // *Strict Pareto Dominance*
    if $\nexists L'' \in \mathrm{Labels}_F(w)$ such that
      $(L''.t \leq t_{\mathrm{new}} \wedge L''.C \supseteq C \wedge (L''.t < t_{\mathrm{new}} \vee L''.C \neq C))$ **then**
      remove all $L'' \in \mathrm{Labels}_F(w)$ such that
        $(t_{\mathrm{new}} \leq L''.t \wedge C \supseteq L''.C \wedge (t_{\mathrm{new}} < L''.t \vee C \neq L''.C))$
      $\mathrm{Labels}_F(w) \leftarrow \mathrm{Labels}_F(w) \cup \{L'\}$
      compute score(L')
      $Q \leftarrow Q \cup \{L'\}$
// *— 5. CLEANUP —*
// *Purge any stored labels that survived Pareto dominance but failed to hit the final country threshold.*
for all $v \in V$ **do**
  for all $L \in \mathrm{Labels}_F(v)$ **do**
    if $|L.C| < \mathrm{MinCountries}$ **then**
      remove L from $\mathrm{Labels}_F(v)$
return Labels_F

Algorithm 10 is a very similar algorithm to Algorithm 9, but in this case we run it only half way to the end of the day (rationale for that explained in Algorithm 12). Since we are doing only until half of the day, in this case we don't do any rollouts (since those run until EOD). Besides, we do not trigger early stopping, because of the record, we expand the entire queue (which we expect to be shorter in this case and hence more feasible).

In the output we keep a dictionnary of routes ending at each specific node, each route containing at least MinCountries.